

第五章 导数相关综合试卷

一、选择题 (Multiple Choice Questions)

1. 函数 $y = \sqrt{x}$ 的导数是 (). What is the derivative of the function $y = \sqrt{x}$?

()

A. $y = \sqrt{x}$

B. $y = \frac{1}{\sqrt{x}}$

C. $y = \frac{1}{2\sqrt{x}}$

D. $y = -\frac{1}{2\sqrt{x}}$

2. 曲线 $y = 2^x$ 在点 $P(3,8)$ 处的切线斜率是 (). What is the slope of the tangent line of the curve $y = 2^x$ at the point $P(3,8)$? ()

A. $7\ln 2$

B. $6\ln 2$

C. $9\ln 2$

D. $8\ln 2$

3. 曲线 $y = \ln x$ 在点 $P(1,0)$ 处的切线方程是 (). What is the equation of the tangent line of the curve $y = \ln x$ at the point $P(1,0)$? ()

A. $y = x - 1$

B. $y = x - e$

C. $y = ex - 1$

D. $y = ex - e$

4. 若物体的运动方程是 $s = 4t^2 + \frac{1}{t}$, 则物体在 $t = 2$ 时的瞬时速度是 (). If the motion equation of an object is $s = 4t^2 + \frac{1}{t}$, what is the instantaneous

velocity of the object at $t = 2$? ()

- A. $\frac{33}{2}$
- B. $\frac{65}{4}$
- C. $\frac{65}{2}$
- D. $\frac{63}{4}$

5. 设函数 $f(x) = \sin x$, 则 $f'(0)$ 等于 (). Let the function $f(x) = \sin x$, then what is $f'(0)$ equal to? ()

- A. 1
- B. -1
- C. 0
- D. None of the above

6. 设函数 $f(x) = \frac{4}{x}$, 则 $f'(1)$ 等于 (). Let the function $f(x) = \frac{4}{x}$, then what is $f'(1)$ equal to? ()

- A. 1
- B. 4
- C. -4
- D. None of the above

7. 函数 $y = 4(2 - x + 3x^2)^2$ 的导数是 (). What is the derivative of the function $y = 4(2 - x + 3x^2)^2$? ()

- A. $8(2 - x + 3x^2)$
- B. $2(-1 + 6x)^2$
- C. $8(2 - x + 3x^2)(6x - 1)$

D. $4(2 - x + 3x^2)(6x - 1)$

8. 如果函数 $y = x^4$, 则 $y^{(5)}$ 等于 (). If the function $y = x^4$, then what is $y^{(5)}$ equal to? ()

A. $4x^3$

B. 0

C. $24x$

D. None of the above

9. 函数 $f(x) = x^3 - 3x - 1$ 的三阶导数是 (). What is the third derivative of the function $f(x) = x^3 - 3x - 1$? ()

A. $3x^2 - 3$

B. $6x$

C. 6

D. 0

10. 函数 $y = \sin x + 2$ 的四阶导数是 (). What is the fourth derivative of the function $y = \sin x + 2$? ()

A. $\sin x$

B. $\cos x$

C. $\sin x + 2$

D. 0

11. 如果 $y = \ln x$, 那么 $y^{(4)}$ 等于 (). If $y = \ln x$, then what is $y^{(4)}$ equal to? ()

A. $\frac{1}{x}$

B. $-\frac{1}{x^2}$

C. $\frac{2}{x^3}$

D. $-\frac{6}{x^4}$

12.如果 $y = y(x)$ 是 $x - y = 0$ 确定的隐函数, $\frac{dy}{dx} = ()$. If $y = y(x)$ is an implicit function determined by $x - y = 0$, then $\frac{dy}{dx} = ()$

A. -1

B. 0

C. 1

D. None of the above

13.如果 $y = y(x)$ 是 $e^y + x^3 + x = 0$ 确定的隐函数, $\frac{dy}{dx} = ()$. If $y = y(x)$ is an implicit function determined by $e^y + x^3 + x = 0$, then $\frac{dy}{dx} = ()$

A. $-\frac{3x^2+1}{e^y}$

B. $\frac{3x^2+1}{e^y}$

C. $\frac{3x^2-1}{e^y}$

D. None of the above

14.如果 $y = y(x)$ 是 $x^2 - xy = 0$ 确定的隐函数, $y' = ()$. If $y = y(x)$ is an implicit function determined by $x^2 - xy = 0$, then $y' = ()$

A. $\frac{x-2y}{x}$

B. $\frac{2x+y}{x}$

C. $\frac{2x-y}{x}$

D. None of the above

15.如果 $y = y(x)$ 是 $\ln y - x = 0$ 确定的隐函数, $y' = ()$. If $y = y(x)$ is an

implicit function determined by $\ln y - x = 0$, then $y' = ()$

A. $\frac{1}{y}$

B. y

C. $\frac{x}{y}$

D. None of the above

16. 如果 $y = y(x)$ 是 $\begin{cases} x = t - 1 \\ y = 2t^3 - 3t^2 + 1 \end{cases}$ 确定的参数函数, $\frac{dy}{dx} = ()$. If $y = y(x)$ is a parametric function determined by $\begin{cases} x = t - 1 \\ y = 2t^3 - 3t^2 + 1 \end{cases}$, then $\frac{dy}{dx} = ()$

A. $2t^2 - 3t$

B. 1

C. $6t^2 - 6t$

D. None of the above

17. 如果 $y = y(x)$ 是 $\begin{cases} x = -\sin t \\ y = 2\cos t \end{cases}$ 确定的参数函数, $\frac{dy}{dx} = ()$. If $y = y(x)$ is a parametric function determined by $\begin{cases} x = -\sin t \\ y = 2\cos t \end{cases}$, then $\frac{dy}{dx} = ()$

A. $-2\tan t$

B. $2\tan t$

C. $2\cot t$

D. None of the above

18. 如果 $y = y(x)$ 是 $\begin{cases} x = \ln t \\ y = e^t \end{cases}$ 确定的参数函数, $y' = ()$. If $y = y(x)$ is a parametric function determined by $\begin{cases} x = \ln t \\ y = e^t \end{cases}$, then $y' = ()$

A. te^t

B. $-te^t$

C. $\frac{e^t}{t}$

D. None of the above

19. 若函数 $f(x) = x^{-2}$, 则 $f'(1)$ 等于 (). If the function $f(x) = x^{-2}$, then what is $f'(1)$ equal to? ()

A. 0

B. -1

C. -2

D. 1

20. 函数 $f(x) = -10$ 的导数是 (). What is the derivative of the function $f(x) = -10$? ()

A. 0

B. Negative number

C. Positive number

D. Uncertain

21. 若 $f(x) = \sqrt[3]{x}$, 则 $f'(1)$ 等于 (). If $f(x) = \sqrt[3]{x}$, then what is $f'(1)$ equal to? ()

A. 0

B. $\frac{1}{3}$

C. 1

D. $\frac{3}{2}$

22. 设函数 $f(x) = -\cos x$, 则 $f'\left(\frac{\pi}{2}\right)$ 等于 (). Let the function $f(x) = -\cos x$,

then what is $f'\left(\frac{\pi}{2}\right)$ equal to? ()

A. 0

B. 1

C. -1

D. None of the above

23. $y = x^n$ 在 $x = 1$ 处切线方程为 $y = -4x + 5$, 则 n 的值为 (). The tangent line equation of $y = x^n$ at $x = 1$ is $y = -4x + 5$, then what is the value of n ? ()

A. 4

B. -4

C. 1

D. -1

24. 如果 $y = x \ln x$, 那么 $y^{(4)}$ 等于 (). If $y = x \ln x$, then what is $y^{(4)}$ equal to? ()

A. $\ln x + 1$

B. $\frac{1}{x}$

C. $-\frac{1}{x^2}$

D. $\frac{2}{x^3}$

25. 如果 $y = y(x)$ 是 $y^2 - xy = 0$ 确定的隐函数, $\frac{dy}{dx} = ()$. If $y = y(x)$ is an implicit function determined by $y^2 - xy = 0$, then $\frac{dy}{dx} = ()$

A. $\frac{2y+x}{y}$

B. $\frac{2y-x}{y}$

C. $\frac{y}{2y-x}$

D. $\frac{y}{2y+x}$

二、填空题 (Fill-in-the-Blank Questions)

1. 函数 $y = \sqrt{x} \cdot 3^x$ 的导数是_____.

What is the derivative of the function $y = \sqrt{x} \cdot 3^x$? _____

2. 曲线 $y = \frac{1}{x^2}$ 在点 $A(2, \frac{1}{4})$ 处的切线斜率是_____；切线方程是_____.

What is the slope of the tangent line of the curve $y = \frac{1}{x^2}$ at the

point $A(2, \frac{1}{4})$? _____; What is the equation of the tangent line?

3. 若物体的运动方程是 $s = 4t^2 + 3$ ，则物体在 $t = 1$ 时的瞬时速度是_____.

If the motion equation of an object is $s = 4t^2 + 3$, what is the

instantaneous velocity of the object at $t = 1$? _____

4. 函数 $y = -\frac{1}{2}x^{\frac{1}{2}}$ ，则 $y' =$ _____.

Given the function $y = -\frac{1}{2}x^{\frac{1}{2}}$, then $y' =$ _____

5. 曲线 $y = \frac{1}{3}x^3$ 在 $x = 1$ 处切线的倾斜角为_____.

What is the inclination angle of the tangent line of the curve $y = \frac{1}{3}x^3$ at

$x = 1$? _____

6. $y = x^3 + \sqrt{x}$ 的导数为_____.

What is the derivative of $y = x^3 + \sqrt{x}$? _____

7. $y = x \sin 2x$ 的导数为_____.

What is the derivative of $y = x \sin 2x$? _____

8. $y = -\tan x$ 的导数为_____.

What is the derivative of $y = -\tan x$? _____

9. $y = -2x^3 + \sqrt{x}$ 的三阶导数为_____.

What is the third derivative of $y = -2x^3 + \sqrt{x}$? _____

10. $y = x \cos x$ 的四阶导数为_____.

What is the fourth derivative of $y = x \cos x$? _____

11. $y = -\tan x$ 的二阶导数为_____.

What is the second derivative of $y = -\tan x$? _____

12. $y = \ln(1+x)$ 的四阶导数为_____.

What is the fourth derivative of $y = \ln(1+x)$? _____

13. 如果 $y = y(x)$ 是 $xy - y^2 = 0$ 确定的隐函数, $\frac{dy}{dx} =$ _____.

If $y = y(x)$ is an implicit function determined by $xy - y^2 = 0$, then $\frac{dy}{dx} =$ _____

14. 如果 $y = y(x)$ 是 $\sin x - y = 0$ 确定的隐函数, $\frac{dy}{dx} =$ _____.

If $y = y(x)$ is an implicit function determined by $\sin x - y = 0$, then $\frac{dy}{dx} =$ _____

15. 如果 $y = y(x)$ 是 $x^3 - xy + y^2 = 0$ 确定的隐函数, $\frac{dy}{dx} =$ _____.

If $y = y(x)$ is an implicit function determined by $x^3 - xy + y^2 = 0$, then $\frac{dy}{dx} =$ _____

16. 如果 $y = y(x)$ 是 $\sin(x-y) + x = y$ 确定的隐函数, $\frac{dy}{dx} =$ _____.

If $y = y(x)$ is an implicit function determined by $\sin(x-y) + x = y$, then $\frac{dy}{dx} =$ _____

17. 如果 $y = y(x)$ 是 $\begin{cases} x = \cos 2t \\ y = \sin 6t \end{cases}$ 确定的参数函数, $\frac{dy}{dx} =$ _____.

If $y = y(x)$ is a parametric function determined by $\begin{cases} x = \cos 2t \\ y = \sin 6t \end{cases}$, then $\frac{dy}{dx} =$ _____

18. 如果 $y = y(x)$ 是 $\begin{cases} x = \cos 2t \\ y = \sin 6t \end{cases}$ 确定的参数函数, $\frac{d^2y}{dx^2} =$ _____.

If $y = y(x)$ is a parametric function determined by $\begin{cases} x = \cos 2t \\ y = \sin 6t \end{cases}$, then $\frac{d^2y}{dx^2} =$ _____

19. 如果 $y = y(x)$ 是 $\begin{cases} x = e^{-t} \\ y = \ln 2t \end{cases}$ 确定的参数函数, $\frac{dy}{dx} =$ _____.

If $y = y(x)$ is a parametric function determined by $\begin{cases} x = e^{-t} \\ y = \ln 2t \end{cases}$, then $\frac{dy}{dx} =$ _____

20. 曲线 $y = x^n$ 在 $x = 2$ 处的导数为 12, 则 n 等于 _____.

The derivative of the curve $y = x^n$ at $x = 2$ is 12, then what is n equal to? _____

21. 函数 $y = \frac{e^x}{x}$, 则 $y' =$ _____.

Given the function $y = \frac{e^x}{x}$, then $y' =$ _____

22. 函数 $y = x \ln x$, 则 $y' =$ _____.

Given the function $y = x \ln x$, then $y' =$ _____

23. $y = 1 - \frac{1}{x^2}$ 的导数为 _____.

What is the derivative of $y = 1 - \frac{1}{x^2}$? _____

24. $y = (x+1)^2(x-1)$ 的导数为 _____; 在 $x = 1$ 处的导数为 _____.

What is the derivative of $y = (x+1)^2(x-1)$? _____; What is the derivative at $x = 1$? _____

25. 如果 $y = \sin^2 x$, 那么 $y^{(4)} =$ _____.

If $y = \sin^2 x$, then $y^{(4)} =$ _____

26. 如果 $y = y(x)$ 是 $e^y \ln x - xy = 0$ 确定的隐函数, $\frac{dy}{dx} =$ _____.

If $y = y(x)$ is an implicit function determined by $e^y \ln x - xy = 0$, then

$\frac{dy}{dx} =$ _____

三、 计算题 (Calculation Questions)

1. 求函数 $y = 2 - 3x^3$ 的导数.

Find the derivative of the function $y = 2 - 3x^3$.

2. 求曲线 $y = \frac{2}{x} - \frac{1}{x^2}$ 在点 $A\left(2, \frac{3}{4}\right)$ 处的切线斜率和切线方程.

Find the slope and the equation of the tangent line of the curve $y = \frac{2}{x} - \frac{1}{x^2}$ at the point $A\left(2, \frac{3}{4}\right)$.

3. 求下列函数的导数. Find the derivatives of the following functions.

(1) $y = \frac{\sin x}{x}$

(2) $y = \cos^2 x - 2x + 1$

4. 如果 $y = \sin^2 x - 3x^2 + 2$, 求 y''' .

If $y = \sin^2 x - 3x^2 + 2$, find y''' .

5. 如果 $y = \sin 3x + 5x^4$, 求 $y^{(4)}$.

If $y = \sin 3x + 5x^4$, find $y^{(4)}$.

6. 如果 $y = y(x)$ 是 $y \ln x - xy^2 = 0$ 确定的隐函数, 求 $\frac{dy}{dx}$.

If $y = y(x)$ is an implicit function determined by $y \ln x - xy^2 = 0$, find $\frac{dy}{dx}$.

7. 如果 $y = y(x)$ 是 $y^4 - 2x^2y + 3x - 5 = 0$ 确定的隐函数, 求 y' 和 $y'|_{x=0}$.

If $y = y(x)$ is an implicit function determined by $y^4 - 2x^2y + 3x - 5 = 0$, find y' and $y'|_{x=0}$.

8. 如果 $y = y(x)$ 是 $\begin{cases} x = -t + 5 \\ y = -\frac{2}{3}t^3 + t \end{cases}$ 确定的参数函数, 求 $\frac{dy}{dx}$ 和 $\frac{d^2y}{dx^2}$.

If $y = y(x)$ is a parametric function determined by $\begin{cases} x = -t + 5 \\ y = -\frac{2}{3}t^3 + t \end{cases}$, find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$.

9. 如果 $y = y(x)$ 是 $\begin{cases} x = \ln(3t + 2) \\ y = e^{2t} \end{cases}$ 确定的参数函数, 求 $y'|_{t=0}$ 和 $y''|_{t=0}$.

If $y = y(x)$ is a parametric function determined by $\begin{cases} x = \ln(3t + 2) \\ y = e^{2t} \end{cases}$, find $y'|_{t=0}$ and $y''|_{t=0}$.

10. 若物体的运动方程是 $s = 8 - 2t + 3t^2$, 则物体在 $t = 2$ 时的瞬时速度是多少?

If the motion equation of an object is $s = 8 - 2t + 3t^2$, what is the instantaneous velocity of the object at $t = 2$?

11. 求指数函数 $y = a^x$ ($a > 0$ 且 $a \neq 1$) 的导数.

Find the derivative of the exponential function $y = a^x$ ($a > 0$ and $a \neq 1$).

12. 求曲线 $y = -\sin x$ 在 $x = \pi$ 处的切线方程.

Find the equation of the tangent line of the curve $y = -\sin x$ at $x = \pi$.

13. 如果 $y = y(x)$ 是 $\begin{cases} x = 2e^{-3t} \\ y = \ln 2t \end{cases}$ 确定的参数函数, 求 $y'|_{t=1}$ 和 $y''|_{t=1}$.

If $y = y(x)$ is a parametric function determined by $\begin{cases} x = 2e^{-3t} \\ y = \ln 2t \end{cases}$, find $y'|_{t=1}$ and $y''|_{t=1}$.

14. 求下列函数的导数. Find the derivatives of the following functions.

(1) $y = \sqrt{x}$

(2) $y = 2\cos x$

(3) $y = 3x^3$